

Chapter 2: Kinematics

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In this Chapter, we will study the motion of particles in space without considering the reasons for the motion; that is what *kinematics* is about. We will introduce the concepts of position, displacement, velocity, and acceleration vectors, and explore their relationships. We will also discuss different types of motion, including linear, circular, spiral, oscillatory, and cycloidal motion. Finally, we will work through some examples to illustrate these concepts. We will stick to Cartesian coordinates for simplicity, but the concepts can be extended to other coordinate systems as well.

1 Notion of a Particle

A *particle* is a body whose size and shape are not important for the problem at hand. In other words, a particle can be treated as a point mass. Its motion is described by its position, velocity, and acceleration as functions of time. Geometrically, a particle is a point in space with an associated mass.

For example, planets can be treated as particles when studying their motion around the Sun, even though they have finite size and shape. However, when studying phenomena such as tidal waves or atmospheric motion, the size and shape of the Earth are important, so the Earth cannot be treated as a particle.

Remark. Since a particle's entire mass is idealized as being concentrated at a single point, its mass density is formally singular at that point. This is an idealization and requires some mathematical care; a rigorous treatment uses the Dirac delta distribution, which will be introduced in the Electricity and Magnetism course.

2 The Position Vector

The position of a particle is described by a vector called the *position vector*:

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}, \tag{1}$$

where $x(t)$, $y(t)$, and $z(t)$ are the Cartesian coordinates of the particle as functions of time, and $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ are the unit vectors in the x -, y -, and z -directions, respectively.

The position vector points from the origin of the coordinate system to the particle. Its magnitude,

$$|\mathbf{r}(t)| = \sqrt{x^2(t) + y^2(t) + z^2(t)}, \quad (2)$$

is the distance of the particle from the origin, while its direction specifies the direction from the origin to the particle.

Because the coordinates can change with time, the position vector $\mathbf{r}(t)$ describes the motion of the particle through space.

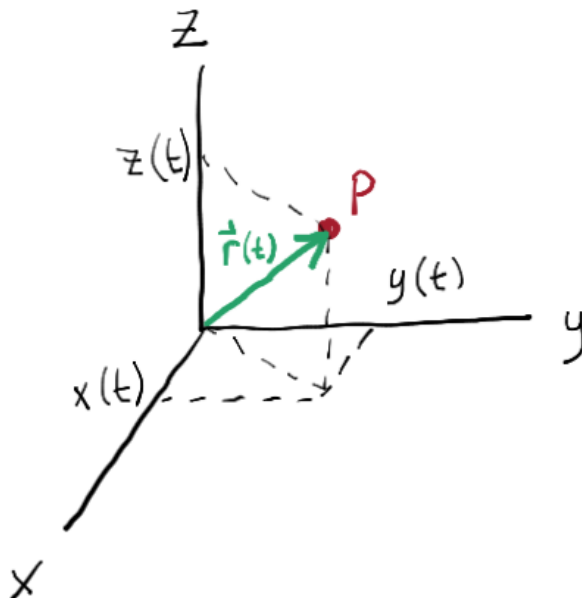


Figure 1: The position vector $\mathbf{r}(t)$ of a particle in a three-dimensional Cartesian coordinate system.

3 The Displacement Vector

Suppose a particle moves from an initial position $\mathbf{r}(t_1)$ to a final position $\mathbf{r}(t_2)$. The *displacement vector* describes the change in the particle's position. It is defined as the difference between the final and initial position vectors:

$$\Delta \mathbf{r} = \mathbf{r}(t_2) - \mathbf{r}(t_1). \quad (3)$$

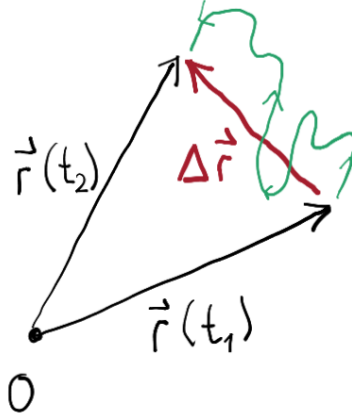


Figure 2: Displacement vector of a particle moving from an initial position $\mathbf{r}(t_1)$ to a final position $\mathbf{r}(t_2)$. The actual path, which is shown by the green curve, taken by the particle is not necessarily a straight line.

The displacement vector points from the initial position to the final position. Its magnitude, $|\Delta \mathbf{r}|$, is the straight-line distance between the initial and final positions. In general, it is not equal to the distance traveled by the particle along its trajectory. The displacement depends only on the initial and final positions, not on the path taken between them.

Geometrically, the relation given in Eq. (3) follows directly from vector subtraction. Since both position vectors originate at the origin, the vector connecting the initial position P_1 to the final position P_2 is equal to $\mathbf{r}(t_2) - \mathbf{r}(t_1)$. Thus, the displacement vector is the vector from the initial position to the final position.

4 The Average Velocity Vector

Velocity describes how the position of a particle changes with time. Suppose the particle moves from an initial position $\mathbf{r}(t_1)$ to a final position $\mathbf{r}(t_2)$ during the time interval $\Delta t = t_2 - t_1$ and let the corresponding displacement be $\Delta \mathbf{r}$. We define *the average velocity vector* as

$$\mathbf{v}_{\text{avg}} = \frac{\Delta \mathbf{r}}{\Delta t}. \quad (4)$$

This quantity describes the displacement per unit time during the interval Δt . Note that the particle may have sped up, slowed down, or even stopped temporarily during the motion from $\mathbf{r}(t_1)$ to $\mathbf{r}(t_2)$. From the initial and final positions alone, we can therefore determine only the particle's motion *on average* over the time interval as given in Eq. (4). In the way the displacement shows where the particle went on average, the average velocity vector shows the direction and magnitude of the particle's motion on average. Consequently, the average velocity vector is in the same direction as the displacement vector, provided $\Delta \mathbf{r} \neq \mathbf{0}$.

Given that the initial and final position vectors be

$$\mathbf{r}(t_1) = x(t_1)\mathbf{i} + y(t_1)\mathbf{j} + z(t_1)\mathbf{k}, \quad (5)$$

$$\mathbf{r}(t_2) = x(t_2)\mathbf{i} + y(t_2)\mathbf{j} + z(t_2)\mathbf{k}, \quad (6)$$

the displacement can be written as

$$\begin{aligned} \Delta \mathbf{r} &= \mathbf{r}(t_2) - \mathbf{r}(t_1) \\ &= [x(t_2) - x(t_1)]\mathbf{i} + [y(t_2) - y(t_1)]\mathbf{j} + [z(t_2) - z(t_1)]\mathbf{k}. \end{aligned} \quad (7)$$

Defining

$$\Delta x = x(t_2) - x(t_1), \quad \Delta y = y(t_2) - y(t_1), \quad \Delta z = z(t_2) - z(t_1), \quad (8)$$

the average velocity can also be written as

$$\mathbf{v}_{\text{avg}} = \frac{\Delta x}{\Delta t}\mathbf{i} + \frac{\Delta y}{\Delta t}\mathbf{j} + \frac{\Delta z}{\Delta t}\mathbf{k}. \quad (9)$$

5 The Instantaneous Velocity Vector

To determine the velocity at a particular instant, we consider the average velocity over an increasingly small time interval around that instant. Mathematically, considering the limit of the average velocity as $\Delta t \rightarrow 0$ leads us to the definition of *instantaneous velocity*

$$\mathbf{v}(t) = \lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{r}}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t}\mathbf{i} + \lim_{\Delta t \rightarrow 0} \frac{\Delta y}{\Delta t}\mathbf{j} + \lim_{\Delta t \rightarrow 0} \frac{\Delta z}{\Delta t}\mathbf{k} \quad (10)$$

where we have the definition of derivative for each term. Hence the acceleration of a particle at time t is defined as

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = \frac{dx}{dt}\mathbf{i} + \frac{dy}{dt}\mathbf{j} + \frac{dz}{dt}\mathbf{k}. \quad (11)$$

Using Newton's notation,

$$\mathbf{v}(t) = \dot{x}(t)\mathbf{i} + \dot{y}(t)\mathbf{j} + \dot{z}(t)\mathbf{k} = v_x(t)\mathbf{i} + v_y(t)\mathbf{j} + v_z(t)\mathbf{k}. \quad (12)$$

The *speed* of the particle is the magnitude of its velocity vector:

$$v(t) = |\mathbf{v}(t)|. \quad (13)$$

We use speed if we are only interested in how fast the particle is moving, regardless of its direction of motion. The speed is always a non-negative quantity, while the velocity can be positive or negative depending on the direction of motion.

Remark. Compared to the average velocity, calculating the instantaneous velocity requires more detailed information about the motion of the particle. In calculating displacement, we only needed two vectors, the initial and final positions. However, in calculating the instantaneous velocity, we need to know $\mathbf{r}(t)$, i.e., all of the position vectors at each instant in time.

6 The Average Acceleration Vector

Acceleration is defined as the change in velocity with respect to time. Following the same reasoning as we did for the average velocity in Sec. 4, we can define the average acceleration vector as follows:

$$\mathbf{a}_{\text{avg}} = \frac{\Delta \mathbf{v}}{\Delta t} \quad (14)$$

where $\Delta \mathbf{v} = \mathbf{v}_2 - \mathbf{v}_1$ is the change in velocity vector, and $\Delta t = t_2 - t_1$ is the time interval over which the change in velocity occurs. The average acceleration vector describes how the velocity of the particle changes on average over the time interval Δt . The average acceleration vector has the same direction as the change in velocity vector. If we are given the initial and final velocity vectors in component form as

$$\mathbf{v}(t_1) = v_x(t_1)\mathbf{i} + v_y(t_1)\mathbf{j} + v_z(t_1)\mathbf{k}$$

$$\mathbf{v}(t_2) = v_x(t_2)\mathbf{i} + v_y(t_2)\mathbf{j} + v_z(t_2)\mathbf{k}$$

Then the change in velocity is

$$\Delta \mathbf{v} = \mathbf{v}(t_2) - \mathbf{v}(t_1) = [v_x(t_2) - v_x(t_1)]\mathbf{i} + [v_y(t_2) - v_y(t_1)]\mathbf{j} + [v_z(t_2) - v_z(t_1)]\mathbf{k}$$

and the average acceleration vector becomes

$$\mathbf{a}_{\text{avg}} = \frac{\Delta v_x}{\Delta t}\mathbf{i} + \frac{\Delta v_y}{\Delta t}\mathbf{j} + \frac{\Delta v_z}{\Delta t}\mathbf{k}$$

7 The Instantaneous Acceleration Vector

If we are interested in the acceleration of the particle at a given time instead of the average acceleration we should consider the behavior of $\mathbf{a}_{\text{avg}}$ when $\Delta t \rightarrow 0$. Therefore,

$$\mathbf{a}(t) = \lim_{\Delta t \rightarrow 0} \mathbf{a}_{\text{avg}} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{v}}{\Delta t} = \frac{d\mathbf{v}}{dt} = \frac{dv_x}{dt} \mathbf{i} + \frac{dv_y}{dt} \mathbf{j} + \frac{dv_z}{dt} \mathbf{k}$$

In Newton's notation

$$\mathbf{a}(t) = \dot{v}_x(t) \mathbf{i} + \dot{v}_y(t) \mathbf{j} + \dot{v}_z(t) \mathbf{k} = a_x(t) \mathbf{i} + a_y(t) \mathbf{j} + a_z(t) \mathbf{k}. \quad (15)$$

where $a_x(t)$, $a_y(t)$, and $a_z(t)$ are the components of the acceleration vector in the x , y , and z directions, respectively.

8 The Trajectory of a Particle

The *trajectory* of a particle is the path followed by the particle in space as time goes on. The *trajectory equation* of a particle is the equation that describes the trajectory of the particle in space. It can be obtained by eliminating the time variable from the components of the position vector $\mathbf{r}(t)$ and connecting the x , y , and z coordinates of the particle.

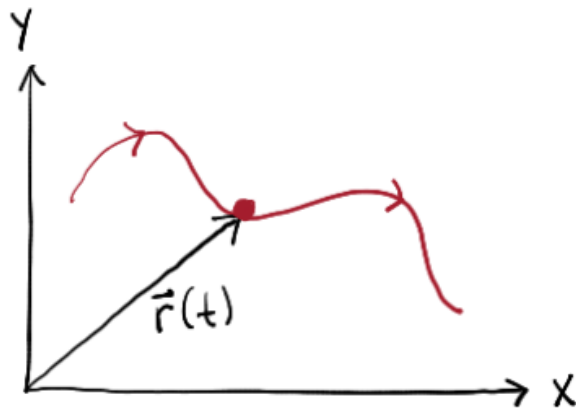


Figure 3: The trajectory of a particle in space is shown in red. The trajectory is the path followed by the particle as time goes on and the equation of the trajectory contains the relationship between the x , y , and z coordinates of the particle.

9 Connecting position, velocity, and acceleration

We have already seen that once we are given the position vector of a particle, $\mathbf{r}(t)$, we can find the velocity and acceleration vectors by taking the first and second derivatives with respect to time, respectively:

$$\mathbf{r}(t) \xrightarrow{\frac{d}{dt}} \mathbf{v}(t) \xrightarrow{\frac{d}{dt}} \mathbf{a}(t) \quad (16)$$

What we really do here is to transform one information into another via a derivative operation. Note that once we know the position vector of a particle, we know everything about the motion of the particle. We can find the velocity and acceleration vectors by taking the first and second derivatives of the position vector with respect to time, respectively.

It is also possible to go the other way around. If we are given the acceleration vector of a particle, $\mathbf{a}(t)$, we can find the velocity and position vectors.

For simplicity assume that the motion takes place on x -axis only and we know $a_x(t)$ as shown in Fig. ??(a), which means that we know how the acceleration of the particle is changing with time. We can find how much the velocity of the particle has changed from time t_1 to time t_2 by integrating the acceleration function with respect to time:

$$\Delta v_x = \int_{t_1}^{t_2} a_x(t) dt = v_x(t_2) - v_x(t_1) \quad (17)$$

Rearranging the above equation, we can write:

$$v_x(t_2) = v_x(t_1) + \int_{t_1}^{t_2} a_x(t) dt \quad (18)$$

This means that if the only information we have is the form of $a_x(t)$, we can find how much the velocity of the particle has changed from time t_1 to time t_2 . However, we still need to know the initial velocity of the particle at time t_1 in order to find the final velocity of the particle at time t_2 .

Remark. If we consider integration as a transformation of information, we can say that we can transform the information about the acceleration of a particle into information about the velocity of the particle. However, we need to know the initial velocity of the particle at time t_1 in order to find the final velocity of the particle at time t_2 .

In terms of indefinite integral, we can write:

$$v_x(t) = \int a_x(t) dt + C \quad (19)$$

where C is a constant that represents the initial velocity of the particle at time $t = 0$. The value of C can be determined if we are given the initial velocity of the particle. Like derivative, the integral is also a linear operator, which means that we can integrate each component of the acceleration vector separately to find the corresponding components of the velocity vector:

$$\mathbf{v}(t) = \int \mathbf{a}(t) dt + \mathbf{C} = \int a_x(t) dt \mathbf{i} + \int a_y(t) dt \mathbf{j} + \int a_z(t) dt \mathbf{k} + \mathbf{C} \quad (20)$$

where $\mathbf{C}$ is a constant vector that represents the initial velocity of the particle at time $t = 0$. The components of $\mathbf{C}$ can be determined if we are given the initial velocity of the particle. Similarly, we can find the position of the particle by integrating the velocity vector:

$$\mathbf{r}(t) = \int \mathbf{v}(t) dt + \mathbf{C} \quad (21)$$

where $\mathbf{C}$ is a constant vector that represents the initial position of the particle at time $t = 0$. The components of $\mathbf{C}$ can be determined if we are given the initial position of the particle.

Summary: If we are given how the position vector of a particle is changing with time, it means that we know everything about the motion of the particle. We can find the velocity and acceleration vectors by taking the first and second derivatives of the position vector with respect to time, respectively. If we are given how the acceleration vector of a particle is changing with time, we can find the velocity and position vectors by taking the indefinite integrals of the acceleration vector with respect to time, respectively. However, we need to know the initial conditions of the particle in order to find its exact position and velocity at any given time.

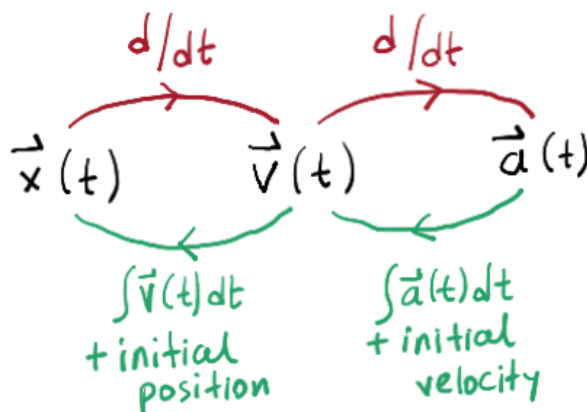


Figure 4: Summary of the relationships between position, velocity, and acceleration vectors.

10 Types of Motion

10.1 Linear Motion

Motion along a straight line is called *linear motion*. In this case, the position vector of the particle can be written as

$$\mathbf{r}(t) = x(t)\mathbf{i}, \quad (22)$$

where the x -axis is chosen along the direction of motion. The velocity and acceleration vectors are also directed along this axis:

$$\mathbf{v}(t) = \dot{x}(t)\mathbf{i}, \quad \mathbf{a}(t) = \ddot{x}(t)\mathbf{i}. \quad (23)$$

When the motion is restricted to a straight line, or in other words, when the motion is one-dimensional, the position, velocity, and acceleration can therefore be described by the scalar quantities

$$x = x(t), \quad v = \dot{x}(t), \quad a = \ddot{x}(t).$$

The sign of each quantity indicates its direction relative to the chosen x -axis. In particular, $v > 0$ means that the particle is moving in the positive x -direction, while $v < 0$ means that it is moving in the negative x -direction. Similarly, $a > 0$ and $a < 0$ indicate acceleration in the positive and negative x -directions, respectively. The magnitude $|v|$ is the speed, while $|a|$ is the magnitude of the acceleration.

10.2 Circular Motion

Circular motion is the motion of a particle along a circular path. For a circular path of constant radius R as shown in Fig. 5, the position vector of the particle can be written as

$$\mathbf{r}(t) = R \cos[\theta(t)] \mathbf{i} + R \sin[\theta(t)] \mathbf{j}. \quad (24)$$

where R is the radius of the circular path and $\theta(t)$ is the angle between the position vector and the positive x -axis. Since the particle moves along the circle, this angle generally changes with time:

$$\theta = \theta(t).$$

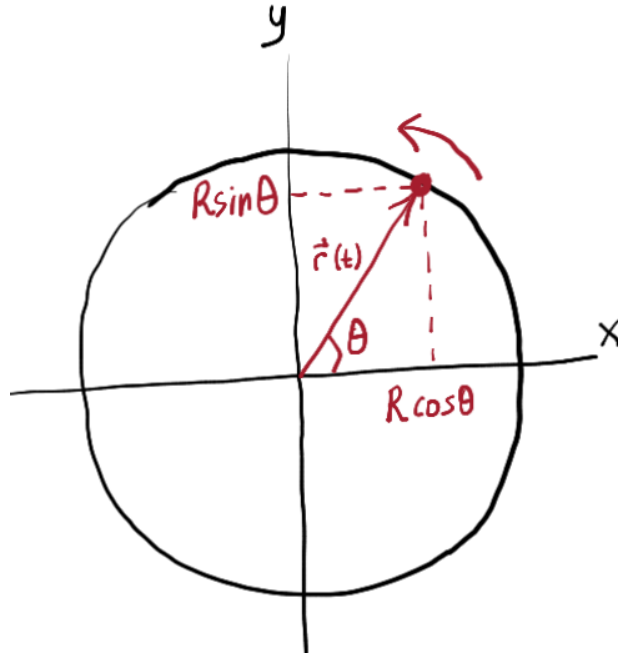


Figure 5: Circular motion described by the position vector $\mathbf{r}(t)$.

The *instantaneous angular velocity* is defined as the rate of change of the angle:

$$\omega(t) = \frac{d\theta}{dt}. \quad (25)$$

In *uniform circular motion*, the speed of the particle is constant. In this case, the angular velocity is also constant, so the angle changes linearly with time:

$$\theta(t) = \omega t + \theta_0, \quad (26)$$

where ω is the constant angular velocity and θ_0 is the angle at $t = 0$.

What is 1 radian?

A radian is a unit of angular measure. An angle of 1 radian is defined as the angle at the center of a circle that subtends an arc whose length is equal to the radius of the circle. Thus, if an angle θ subtends an arc of length s on a circle of radius R , then, when θ is measured in radians,

$$\theta = \frac{s}{R}. \quad (27)$$

For a complete circle, the arc length is the circumference $2\pi R$. Therefore,

$$2\pi \text{ rad} = 360^\circ.$$

The definition of the radian will be used repeatedly in the following sections, so it is important to understand this relationship between angular displacement and arc length. The following figure illustrates the definition of one radian:

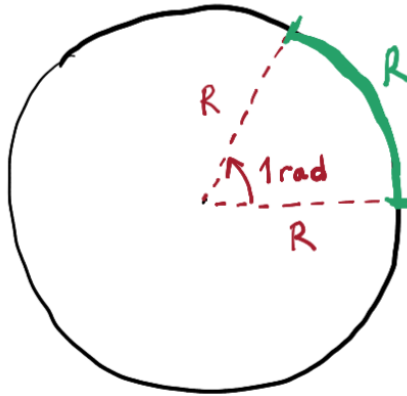


Figure 6: Definition of one radian: an angle of 1 radian subtends an arc whose length equals the radius of the circle.

We can obtain the velocity vector of a particle moving in a circular path by differentiating the position vector with respect to time:

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = \frac{d}{dt} [R \cos \theta \mathbf{i} + R \sin \theta \mathbf{j}] = -R\dot{\theta} \sin \theta \mathbf{i} + R\dot{\theta} \cos \theta \mathbf{j}. \quad (28)$$

Since $\dot{\theta} = \omega$ for uniform circular motion,

$$\mathbf{v}(t) = R\omega [-\sin \theta \mathbf{i} + \cos \theta \mathbf{j}]. \quad (29)$$

The velocity vector is tangent to the circular path and therefore perpendicular to the position vector. Its magnitude is

$$|\mathbf{v}| = R|\omega|.$$

Thus, for uniform circular motion, the speed is constant and is related to the angular velocity by

$$v = R|\omega|.$$

Similarly, we can find the acceleration vector by differentiating the velocity vector with respect to time:

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = \frac{d}{dt} [R\omega (-\sin \theta \mathbf{i} + \cos \theta \mathbf{j})] = -R\omega^2 \cos \theta \mathbf{i} - R\omega^2 \sin \theta \mathbf{j}. \quad (30)$$

Thus, for uniform circular motion, the acceleration vector points towards the center of the circular path and is therefore directed radially inward. Its magnitude is

$$a = R|\omega|^2 \quad (31)$$

We can also express the acceleration in terms of the speed v :

$$a = \frac{R^2}{R} \left(\frac{d\theta}{dt} \right)^2 = \frac{1}{R} \left(\frac{ds}{dt} \right)^2 = \frac{v^2}{R} \quad (32)$$

10.3 Spiral Motion

It is a composite motion of linear and circular motions. The particle moves along a spiral path, which can be described by a combination of linear and circular motion equations. The position vector of a particle moving in a spiral path can be expressed as:

$$\mathbf{r}(t) = R \cos[\theta(t)] \mathbf{i} + R \sin[\theta(t)] \mathbf{j} + z(t) \mathbf{k}$$

which describes a circular motion in the xy -plane with radius R and a linear motion along the z -axis. The angle $\theta(t)$ and the height $z(t)$ can be functions of time, allowing for various types of spiral motion, such as helical or conical spirals.

10.4 Oscillatory Motion

Oscillations are periodic motions; they repeat themselves after a certain interval of time. Complex numbers together with the exponential function provide a powerful and elegant framework for describing and analyzing oscillatory phenomena.

Complex Numbers

Mathematicians in the 16th century discovered that cubic formulas sometimes yield three real solutions while still forcing square roots of negative numbers to appear in the intermediate algebra. Bombelli (1572) was the first to state systematic rules for operating with $\sqrt{-1}$, making these calculations consistent and allowing the real results to be recovered, marking the first systematic use of complex numbers. Descartes later coined the term “imaginary” for these numbers, reflecting the skepticism of the time regarding their existence, and Euler adopted the notation i for $\sqrt{-1}$, making the calculations routine; this notation has since become standard in mathematics.

Complex numbers are numbers that can be expressed in the form

$$z = x + iy, \quad (33)$$

where x and y are real numbers and i is the imaginary unit satisfying $i^2 = -1$. We call them “numbers” because they obey the same algebraic rules as real numbers, with the additional rule that $i^2 = -1$, allowing addition, subtraction, multiplication, and division.

The real part of z is denoted $\text{Re}(z) = x$ and the imaginary part $\text{Im}(z) = y$. Complex numbers can be represented graphically on the *complex plane*, where the horizontal axis represents the real part and the vertical axis the imaginary part, as shown in Fig. 7.

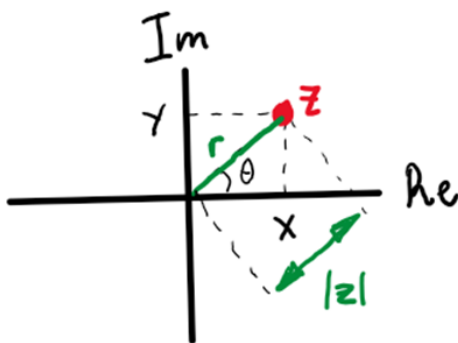


Figure 7: The complex plane.

By the Pythagorean theorem, the magnitude (or modulus) of $z = x + iy$ is

$$|z| = \sqrt{x^2 + y^2}, \quad (34)$$

the distance from the origin to the point (x, y) in the complex plane. The argument (or angle) of z is the angle θ that the line from the origin to z makes with the positive real axis. For $x > 0$ it reduces to

$$\theta = \tan^{-1}\left(\frac{y}{x}\right); \quad (35)$$

in general the correct quadrant must be taken into account, and the two-argument function $\text{atan2}(y, x)$ returns the correct angle for all $(x, y) \neq (0, 0)$.

It is often useful to write z in *polar form* (see Fig. 7), which expresses z in terms of its magnitude and argument:

$$z = r(\cos \theta + i \sin \theta), \quad (36)$$

where $r = |z|$ is the magnitude and $\theta = \arg(z)$ is the argument.

The connection between complex numbers, circular motion and oscillations

When r is held constant and θ changes linearly with time, z traces a circle in the complex plane, representing circular motion. Setting $\theta = \omega t + \phi$, where ω is the angular velocity (in units $[\text{T}^{-1}]$) as we defined it in Sec. 10.2, and ϕ is the phase constant, gives

$$z(t) = r[\cos(\omega t + \phi) + i \sin(\omega t + \phi)]. \quad (37)$$

If we take the real part of $z(t)$, we obtain the familiar equation for simple harmonic oscillation which will be examined in more detail in Sec. ??:

$$x(t) = r \cos(\omega t + \phi), \quad (38)$$

which describes the motion of a particle oscillating back and forth along a line. It is customary to call ω in this case the *angular frequency* since it refers to the rate of change of the phase of an oscillation. However, both angular velocity and angular frequency have the same units and the same numerical value. The imaginary part

$$y(t) = r \sin(\omega t + \phi) \quad (39)$$

describes a particle oscillating along a line perpendicular to the x -axis (imaginary axis). The combination of these two perpendicular oscillations results in circular motion, as the particle moves around the circle in the complex plane. The angular velocity ω determines how fast the particle moves around the circle, while the phase constant ϕ determines the initial position of the particle on the circle at $t = 0$.

The Exponential Function

The exponential function e^x describes growth or decay processes, as shown in Fig. 8.

It is defined by its Taylor series:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \quad (40)$$

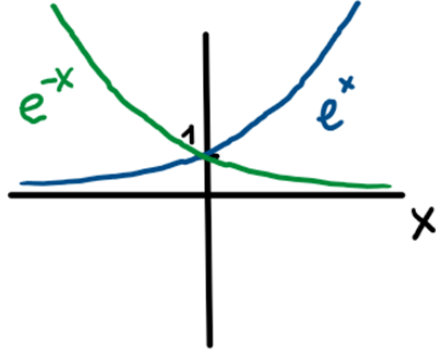


Figure 8: The exponential function e^x : growth for positive x , decay for negative x .

A key property is that e^x is the *unique* function satisfying $f' = f$ with $f(0) = 1$, meaning its rate of change at every point equals its value there:

$$\frac{d}{dx} e^x = e^x. \quad (41)$$

More generally, for a constant rate λ , where λ is a real number:

$$\frac{d}{dt} e^{\lambda t} = \lambda e^{\lambda t}, \quad (42)$$

so the velocity of a particle whose position is $x(t) = e^{\lambda t}$ is always proportional to its position by the factor λ : exponential growth for $\lambda > 0$ and exponential decay for $\lambda < 0$.

Complex Exponential Function

Substituting $x = i\theta$ (with θ real) into the Taylor series of e^x and using $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, ... yields

$$e^{i\theta} = 1 + i\theta - \frac{\theta^2}{2!} - \frac{i\theta^3}{3!} + \frac{\theta^4}{4!} + \dots \quad (43)$$

Separating real and imaginary parts:

$$e^{i\theta} = \underbrace{\left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots\right)}_{\cos \theta} + i \underbrace{\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots\right)}_{\sin \theta}, \quad (44)$$

which is *Euler's formula*:

$$e^{i\theta} = \cos \theta + i \sin \theta. \quad (45)$$

Applying this to the polar form, the time-dependent complex number becomes simply

$$z(t) = r e^{i(\omega t + \phi)}. \quad (46)$$

While a cosine or sine alone can describe a single oscillation, the complex exponential handles multiple oscillations, phase relationships, and operations such as differentiation and integration more efficiently, making it the preferred tool in signal processing and wave mechanics.

Compared to Eq. (42), replacing the real constant λ with the imaginary constant $i\omega$ gives

$$\frac{d}{dt} e^{i\omega t} = i\omega e^{i\omega t}, \quad (47)$$

so the velocity of a particle whose position is $x(t) = e^{i\omega t}$ is always proportional to its position by the factor $i\omega$. The factor i indicates that the velocity is perpendicular to the position, as in circular motion. The magnitude of the velocity is $|\mathbf{v}| = \omega|x|$, as we found in Sec. 10.2.

The coefficient of the time parameter t in the exponent can be chosen as a general complex number. In this way, we can describe a combination of oscillatory and exponential growth or decay behavior. For example, if we let $\lambda = \alpha + i\omega$, where α and ω are real numbers, then

$$\frac{d}{dt} e^{(\alpha+i\omega)t} = (\alpha + i\omega) e^{(\alpha+i\omega)t}, \quad (48)$$

so the velocity of a particle whose position is $x(t) = e^{(\alpha+i\omega)t}$ is always proportional to its position by the factor $\alpha+i\omega$. The real part α indicates that the particle's speed is increasing or decreasing exponentially, while the imaginary part $i\omega$ indicates that the particle is oscillating. The magnitude of the velocity is

$$|\mathbf{v}| = \sqrt{\alpha^2 + \omega^2} |x|, \quad (49)$$

which combines the effects of exponential growth or decay and oscillation.

10.5 Cycloidal Motion — *Optional*

It is the path traced by a point on the circumference of a circle as it rolls along straight line without slipping. Imagine drawing a chalk dot on a bicycle wheel and then rolling the wheel along a flat surface. The path traced by the chalk dot is a cycloid.

The position vector of a particle moving along a cycloidal path can be expressed as:

$$\mathbf{r}(t) = R(t - \sin t)\mathbf{i} + R(1 - \cos t)\mathbf{j}$$

11 Worked Examples

11.1 Collision of Two Particles

Two particles are moving on a plane according to the following position vectors:

$$\mathbf{r}_1(t) = t\mathbf{i} + t\mathbf{j}$$

$$\mathbf{r}_2(t) = (-t + 4)\mathbf{i} + (2t - 2)\mathbf{j}$$

What time t do the two particles collide? What is the position vector of the collision point?

Solution 1: The two particles collide when their position vectors are equal:

$$\mathbf{r}_1(t) = \mathbf{r}_2(t)$$

This gives us two equations:

$$t = -t + 4 \quad (\text{i})$$

$$t = 2t - 2 \quad (\text{ii})$$

From equation (i), we get:

$$2t = 4 \implies t = 2$$

From equation (ii), we get:

$$t = 2t - 2 \implies t = 2$$

Both equations give us the same time $t = 2$. Now we can find the position vector of the collision point by substituting $t = 2$ into either $\mathbf{r}_1(t)$ or $\mathbf{r}_2(t)$:

$$\mathbf{r}_1(2) = 2\mathbf{i} + 2\mathbf{j} = 2\mathbf{i} + 2\mathbf{j}$$

Solution 2: Define a *separation vector* $\Delta\mathbf{r}$:

$$\Delta\mathbf{r} = \mathbf{r}_2(t) - \mathbf{r}_1(t)$$

The two particles collide when $|\Delta\mathbf{r}| = 0$:

$$|(-2t + 4)\mathbf{i} + (t - 2)\mathbf{j}| = 0$$

This is equivalent to:

$$\sqrt{(-2t + 4)^2 + (t - 2)^2} = 0$$

Squaring both sides:

$$(-2t + 4)^2 + (t - 2)^2 = 0$$

Expanding:

$$4t^2 - 16t + 16 + t^2 - 4t + 4 = 0$$

Combining like terms:

$$5t^2 - 20t + 20 = 0$$

Dividing by 5:

$$t^2 - 4t + 4 = 0$$

This factors as:

$$(t - 2)^2 = 0$$

So $t = 2$.

11.2 Displacement, Average Velocity, Instantaneous Velocity, and Collision

Two particles move in the xy -plane according to

$$\mathbf{r}_1(t) = (t^2 + 1)\mathbf{i} + 2t\mathbf{j},$$

$$\mathbf{r}_2(t) = (5 - t)\mathbf{i} + (t^2 - 1)\mathbf{j}.$$

1. Find the position of each particle at $t = 0$ and $t = 2$.
2. Find the displacement vectors of both particles between $t = 0$ and $t = 2$.
3. Find their average velocity vectors over this time interval.
4. Find $\mathbf{v}_1(t)$ and $\mathbf{v}_2(t)$.
5. Determine whether the particles collide. If they do, find the collision time and collision position.

Solution:

(a) Positions at $t = 0$ and $t = 2$

For particle 1,

$$\mathbf{r}_1(0) = (0^2 + 1)\mathbf{i} + 2(0)\mathbf{j} = \mathbf{i},$$

and

$$\mathbf{r}_1(2) = (2^2 + 1)\mathbf{i} + 2(2)\mathbf{j} = 5\mathbf{i} + 4\mathbf{j}.$$

For particle 2,

$$\mathbf{r}_2(0) = (5 - 0)\mathbf{i} + (0^2 - 1)\mathbf{j} = 5\mathbf{i} - \mathbf{j},$$

and

$$\mathbf{r}_2(2) = (5 - 2)\mathbf{i} + (2^2 - 1)\mathbf{j} = 3\mathbf{i} + 3\mathbf{j}.$$

Therefore,

$$\mathbf{r}_1(0) = \mathbf{i}, \quad \mathbf{r}_1(2) = 5\mathbf{i} + 4\mathbf{j}$$

and

$$\mathbf{r}_2(0) = 5\mathbf{i} - \mathbf{j}, \quad \mathbf{r}_2(2) = 3\mathbf{i} + 3\mathbf{j}.$$

(b) Displacement vectors

The displacement is

$$\Delta \mathbf{r} = \mathbf{r}(t_2) - \mathbf{r}(t_1).$$

Thus, for particle 1,

$$\Delta \mathbf{r}_1 = \mathbf{r}_1(2) - \mathbf{r}_1(0) = (5\mathbf{i} + 4\mathbf{j}) - \mathbf{i},$$

so

$$\Delta \mathbf{r}_1 = 4\mathbf{i} + 4\mathbf{j}.$$

For particle 2,

$$\Delta \mathbf{r}_2 = \mathbf{r}_2(2) - \mathbf{r}_2(0) = (3\mathbf{i} + 3\mathbf{j}) - (5\mathbf{i} - \mathbf{j}),$$

hence

$$\Delta \mathbf{r}_2 = -2\mathbf{i} + 4\mathbf{j}.$$

(c) Average velocity vectors

The average velocity is

$$\mathbf{v}_{\text{avg}} = \frac{\Delta \mathbf{r}}{\Delta t}.$$

Since

$$\Delta t = 2 - 0 = 2,$$

we obtain

$$\mathbf{v}_{\text{avg},1} = \frac{4\mathbf{i} + 4\mathbf{j}}{2} = 2\mathbf{i} + 2\mathbf{j},$$

and therefore

$$\boxed{\mathbf{v}_{\text{avg},1} = 2\mathbf{i} + 2\mathbf{j}.}$$

Similarly,

$$\mathbf{v}_{\text{avg},2} = \frac{-2\mathbf{i} + 4\mathbf{j}}{2} = -\mathbf{i} + 2\mathbf{j},$$

so

$$\boxed{\mathbf{v}_{\text{avg},2} = -\mathbf{i} + 2\mathbf{j}.}$$

(d) Instantaneous velocity

The instantaneous velocity is obtained from

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}.$$

For particle 1,

$$\mathbf{v}_1(t) = \frac{d}{dt} [(t^2 + 1)\mathbf{i} + 2t\mathbf{j}],$$

giving

$$\boxed{\mathbf{v}_1(t) = 2t\mathbf{i} + 2\mathbf{j}.}$$

For particle 2,

$$\mathbf{v}_2(t) = \frac{d}{dt} [(5 - t)\mathbf{i} + (t^2 - 1)\mathbf{j}],$$

so

$$\boxed{\mathbf{v}_2(t) = -\mathbf{i} + 2t\mathbf{j}.}$$

(e) Collision

The particles collide when their position vectors are equal at the same time:

$$\mathbf{r}_1(t) = \mathbf{r}_2(t).$$

This gives the two equations

$$t^2 + 1 = 5 - t,$$

and

$$2t = t^2 - 1.$$

The first equation gives

$$t^2 + t - 4 = 0,$$

so

$$t = \frac{-1 \pm \sqrt{17}}{2}.$$

The second equation gives

$$t^2 - 2t - 1 = 0,$$

so

$$t = 1 \pm \sqrt{2}.$$

There is no time that satisfies both equations simultaneously. Therefore, the particles do not collide.

The two particles never collide.

This illustrates that equality of one coordinate is not sufficient for a collision; all spatial coordinates must be equal at the same instant.

11.3 Velocity, Acceleration, and the Interpretation of Signs

A particle moves along the x -axis according to

$$x(t) = t^3 - 6t^2 + 9t.$$

1. Find $v(t)$ and $a(t)$.
2. Determine the times at which the particle is at rest.
3. For each time interval separated by these instants, determine whether the particle is moving in the $+x$ or $-x$ direction.
4. Determine where the particle is speeding up and where it is slowing down.
5. Find the total distance traveled during $0 \leq t \leq 4$.
6. Compare the total distance traveled with the magnitude of the net displacement.

Solution:

(a) Velocity and acceleration

The velocity is the derivative of position:

$$v(t) = \frac{dx}{dt} = 3t^2 - 12t + 9.$$

Factoring,

$$v(t) = 3(t^2 - 4t + 3) = 3(t - 1)(t - 3).$$

Therefore,

$$\boxed{v(t) = 3(t - 1)(t - 3).}$$

The acceleration is

$$a(t) = \frac{dv}{dt} = 6t - 12,$$

so

$$\boxed{a(t) = 6(t - 2).}$$

(b) Times at which the particle is at rest

The particle is at rest when

$$v(t) = 0.$$

Therefore,

$$3(t - 1)(t - 3) = 0,$$

which gives

$$\boxed{t = 1, \quad t = 3.}$$

(c) Direction of motion

The sign of $v(t)$ determines the direction of motion.

For $0 < t < 1$,

$$(t - 1) < 0, \quad (t - 3) < 0,$$

so

$$v(t) > 0.$$

Thus the particle moves in the $+x$ direction.

For $1 < t < 3$,

$$(t - 1) > 0, \quad (t - 3) < 0,$$

so

$$v(t) < 0.$$

Thus the particle moves in the $-x$ direction.

For $t > 3$,

$$(t - 1) > 0, \quad (t - 3) > 0,$$

so

$$v(t) > 0.$$

Thus the particle again moves in the $+x$ direction.

Therefore,

Time interval	Direction
$0 < t < 1$	$+x$
$1 < t < 3$	$-x$
$t > 3$	$+x$

(d) Speeding up and slowing down

A particle speeds up when the velocity and acceleration have the same sign, and slows down when they have opposite signs.

We have

$$a(t) = 6(t - 2).$$

Hence,

$$a(t) < 0 \quad \text{for } t < 2,$$

and

$$a(t) > 0 \quad \text{for } t > 2.$$

Combining this with the sign of $v(t)$:

Interval	v	a	Behavior
$0 < t < 1$	$+$	$-$	slowing down
$1 < t < 2$	$-$	$-$	speeding up
$2 < t < 3$	$-$	$+$	slowing down
$t > 3$	$+$	$+$	speeding up

Thus,

slowing down: $0 < t < 1, \quad 2 < t < 3,$
speeding up: $1 < t < 2, \quad t > 3.$

(e) Total distance traveled for $0 \leq t \leq 4$

Because the particle changes direction at $t = 1$ and $t = 3$, we must divide the motion into separate intervals.

First calculate the positions:

$$x(0) = 0,$$

$$x(1) = 1 - 6 + 9 = 4,$$

$$x(3) = 27 - 54 + 27 = 0,$$

$$x(4) = 64 - 96 + 36 = 4.$$

Therefore, the distances traveled are

$$d_1 = |x(1) - x(0)| = |4 - 0| = 4,$$

$$d_2 = |x(3) - x(1)| = |0 - 4| = 4,$$

$$d_3 = |x(4) - x(3)| = |4 - 0| = 4.$$

Hence the total distance is

$$\boxed{d_{\text{total}} = 4 + 4 + 4 = 12.}$$

(f) Distance versus displacement

The net displacement is

$$\Delta x = x(4) - x(0) = 4 - 0 = 4.$$

Its magnitude is therefore

$$|\Delta x| = 4.$$

Thus,

total distance = 12, displacement = 4.

The difference occurs because the particle changes direction during its motion.

11.4 Reconstructing Position and Velocity from Acceleration

A particle moves in one dimension with acceleration

$$a(t) = 6t - 4.$$

At $t = 0$,

$$x(0) = 2, \quad v(0) = -3.$$

1. Find $v(t)$.
2. Find $x(t)$.
3. Determine the time at which the particle changes direction.
4. Find the position of the particle at that instant.
5. Verify the result by differentiating $x(t)$.

Solution:

(a) Finding the velocity

Since

$$a(t) = \frac{dv}{dt},$$

we integrate:

$$v(t) = \int (6t - 4) dt.$$

Therefore,

$$v(t) = 3t^2 - 4t + C_1.$$

Using the initial condition $v(0) = -3$,

$$-3 = C_1.$$

Hence,

$$\boxed{v(t) = 3t^2 - 4t - 3.}$$

(b) Finding the position

Since

$$v(t) = \frac{dx}{dt},$$

we integrate again:

$$x(t) = \int (3t^2 - 4t - 3) dt.$$

Thus,

$$x(t) = t^3 - 2t^2 - 3t + C_2.$$

Using $x(0) = 2$,

$$2 = C_2.$$

Therefore,

$$\boxed{x(t) = t^3 - 2t^2 - 3t + 2.}$$

This demonstrates why the initial conditions are necessary: one integration introduces one unknown constant, and a second integration introduces another.

(c) Time at which the particle changes direction

The particle changes direction when its velocity changes sign. First find when it is at rest:

$$v(t) = 0.$$

Thus,

$$3t^2 - 4t - 3 = 0.$$

Using the quadratic formula,

$$t = \frac{4 \pm \sqrt{16 + 36}}{6} = \frac{4 \pm 2\sqrt{13}}{6}.$$

Therefore,

$$t = \frac{2 \pm \sqrt{13}}{3}.$$

Since physical time in this problem is taken to begin at $t = 0$, the relevant solution is

$$t_{\text{turn}} = \frac{2 + \sqrt{13}}{3}.$$

(d) Position at the turning point

Substituting

$$t_{\text{turn}} = \frac{2 + \sqrt{13}}{3}$$

into

$$x(t) = t^3 - 2t^2 - 3t + 2$$

gives the turning-point position

$$x_{\text{turn}} = -\frac{52 + 26\sqrt{13}}{27}.$$

Numerically,

$$x_{\text{turn}} \approx -5.412.$$

(e) Verification

Differentiating the position function,

$$x(t) = t^3 - 2t^2 - 3t + 2,$$

gives

$$\frac{dx}{dt} = 3t^2 - 4t - 3 = v(t),$$

and differentiating once more,

$$\frac{d^2x}{dt^2} = 6t - 4 = a(t).$$

Therefore,

$$\boxed{\frac{dx}{dt} = v(t), \quad \frac{d^2x}{dt^2} = a(t),}$$

as required.

11.5 Uniform Circular Motion

A particle moves on a circle of radius R according to

$$\mathbf{r}(t) = R \cos(\omega t) \mathbf{i} + R \sin(\omega t) \mathbf{j}.$$

1. Find $\mathbf{v}(t)$.
2. Find the speed $|\mathbf{v}(t)|$.
3. Find $\mathbf{a}(t)$.
4. Show explicitly that

$$\mathbf{r}(t) \cdot \mathbf{v}(t) = 0.$$

5. Show that the acceleration always points toward the center of the circle.
6. Find the magnitude of the acceleration in terms of R and ω .
7. Explain geometrically why the velocity is tangent to the trajectory while the acceleration is directed toward the center.

Solution:

(a) Velocity

The velocity is

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}.$$

Therefore,

$$\mathbf{v}(t) = -R\omega \sin(\omega t)\mathbf{i} + R\omega \cos(\omega t)\mathbf{j}.$$

Hence,

$$\boxed{\mathbf{v}(t) = -R\omega \sin(\omega t)\mathbf{i} + R\omega \cos(\omega t)\mathbf{j}.}$$

(b) Speed

The speed is the magnitude of the velocity:

$$|\mathbf{v}(t)| = \sqrt{R^2\omega^2 \sin^2(\omega t) + R^2\omega^2 \cos^2(\omega t)}.$$

Using

$$\sin^2 \theta + \cos^2 \theta = 1,$$

we obtain

$$|\mathbf{v}(t)| = R|\omega|.$$

For $\omega > 0$,

$$\boxed{v = R\omega.}$$

Thus the speed is constant.

(c) Acceleration

Differentiate the velocity:

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt}.$$

Hence,

$$\mathbf{a}(t) = -R\omega^2 \cos(\omega t)\mathbf{i} - R\omega^2 \sin(\omega t)\mathbf{j}.$$

Factorizing,

$$\boxed{\mathbf{a}(t) = -\omega^2 [R \cos(\omega t)\mathbf{i} + R \sin(\omega t)\mathbf{j}].}$$

Since the expression in brackets is precisely $\mathbf{r}(t)$,

$$\boxed{\mathbf{a}(t) = -\omega^2 \mathbf{r}(t).}$$

(d) Orthogonality of position and velocity

Using the expressions above,

$$\mathbf{r}(t) \cdot \mathbf{v}(t) = [R \cos(\omega t)\mathbf{i} + R \sin(\omega t)\mathbf{j}] \cdot [-R\omega \sin(\omega t)\mathbf{i} + R\omega \cos(\omega t)\mathbf{j}] \quad (50)$$

$$= -R^2\omega \cos(\omega t) \sin(\omega t) + R^2\omega \sin(\omega t) \cos(\omega t) \quad (51)$$

$$= 0. \quad (52)$$

Therefore,

$$\boxed{\mathbf{r}(t) \perp \mathbf{v}(t).}$$

Since the position vector points from the center to the particle, the velocity is perpendicular to the radius and therefore tangent to the circle.

(e) Direction of the acceleration

We found that

$$\mathbf{a}(t) = -\omega^2 \mathbf{r}(t).$$

The factor $-\omega^2$ is negative, so $\mathbf{a}(t)$ points in the direction opposite to $\mathbf{r}(t)$.

Since $\mathbf{r}(t)$ points from the center toward the particle, the acceleration points from the particle toward the center.

Thus,

$$\boxed{\text{the acceleration is always directed toward the center.}}$$

(f) Magnitude of the acceleration

Since

$$\mathbf{a}(t) = -\omega^2 \mathbf{r}(t),$$

we have

$$|\mathbf{a}(t)| = \omega^2 |\mathbf{r}(t)|.$$

Because

$$|\mathbf{r}(t)| = R,$$

it follows that

$$a = R\omega^2.$$

Using $v = R\omega$, this can also be written as

$$a = \frac{v^2}{R}.$$

(g) Geometrical interpretation

The particle remains at a fixed distance R from the center. Therefore, its position vector continuously rotates.

The velocity is the rate of change of the position vector. An infinitesimal change in the position vector along a circular path is tangent to the circle, so the instantaneous velocity is tangent to the trajectory.

The velocity vector itself changes direction continuously. Its change is directed toward the center of the circle, producing an inward acceleration.

Thus,

$$\mathbf{v} \text{ is tangent to the trajectory,} \quad \mathbf{a} \text{ points toward the center.}$$

11.6 Oscillatory Motion and the Complex Exponential

Consider the complex function

$$z(t) = Ae^{i(\omega t + \phi)},$$

where A , ω , and ϕ are constants.

1. Using Euler's formula, write $z(t)$ in terms of its real and imaginary parts.
2. What trajectory does the corresponding point follow in the complex plane?
3. Calculate $\dot{z}(t)$ and $\ddot{z}(t)$.
4. Show that

$$\ddot{z} + \omega^2 z = 0.$$

5. What are the real and imaginary parts of this equation?
6. Explain why the complex exponential is useful for describing oscillatory motion.

Solution:

(a) Real and imaginary parts

Using Euler's formula,

$$e^{i\theta} = \cos \theta + i \sin \theta,$$

we obtain

$$z(t) = A [\cos(\omega t + \phi) + i \sin(\omega t + \phi)].$$

Therefore,

$$\boxed{\operatorname{Re}[z(t)] = A \cos(\omega t + \phi)}$$

and

$$\boxed{\operatorname{Im}[z(t)] = A \sin(\omega t + \phi).}$$

Thus, the same complex function contains both sine and cosine oscillations.

(b) Trajectory in the complex plane

The real and imaginary coordinates are

$$x(t) = A \cos(\omega t + \phi),$$

$$y(t) = A \sin(\omega t + \phi).$$

Squaring and adding,

$$x^2 + y^2 = A^2 \cos^2(\omega t + \phi) + A^2 \sin^2(\omega t + \phi).$$

Hence,

$$x^2 + y^2 = A^2.$$

Therefore, the point follows a circle of radius A centered at the origin:

$$\boxed{x^2 + y^2 = A^2.}$$

(c) First and second derivatives

Differentiating,

$$\dot{z}(t) = i\omega A e^{i(\omega t + \phi)}.$$

Therefore,

$$\boxed{\dot{z}(t) = i\omega z(t).}$$

Differentiating again,

$$\ddot{z}(t) = (i\omega)^2 A e^{i(\omega t + \phi)}.$$

Since

$$i^2 = -1,$$

we have

$$\boxed{\ddot{z}(t) = -\omega^2 z(t).}$$

(d) Differential equation

From the previous result,

$$\ddot{z} = -\omega^2 z.$$

Therefore,

$$\boxed{\ddot{z} + \omega^2 z = 0.}$$

Thus the complex exponential automatically satisfies the differential equation of simple harmonic motion.

(e) Real and imaginary parts

Since

$$z(t) = x(t) + iy(t),$$

we have

$$\ddot{z} + \omega^2 z = (\ddot{x} + \omega^2 x) + i(\ddot{y} + \omega^2 y).$$

For this expression to vanish, both the real and imaginary parts must vanish:

$$\boxed{\ddot{x} + \omega^2 x = 0}$$

and

$$\boxed{\ddot{y} + \omega^2 y = 0.}$$

Thus both the real and imaginary components independently describe simple harmonic motion.

(f) Why use complex exponentials?

The complex exponential provides a compact representation of two real oscillatory functions:

$$e^{i\omega t} = \cos(\omega t) + i \sin(\omega t).$$

Differentiation becomes particularly simple:

$$\frac{d}{dt}e^{i\omega t} = i\omega e^{i\omega t}.$$

Similarly,

$$\frac{d^2}{dt^2}e^{i\omega t} = -\omega^2 e^{i\omega t}.$$

Thus, instead of carrying separate sine and cosine functions through every calculation, we can perform the calculation using a single complex exponential and take the real or imaginary part at the end.

The complex representation therefore provides a convenient mathematical tool for analyzing oscillatory motion, while the underlying physical motion can remain completely real.